4.1 Theoretical Background

4.1.1 About Autopsy Software:

The Autopsy is a digital forensics platform and graphical interface to The Sleuth Kit and other digital forensics tools. It is used by law enforcement, military and corporate examiners to investigate what happened on a computer. You can even use it to recover photos from your camera's memory card.

4.1.2 Autopsy supports disk images in the following formats:

Raw Single (For example: *.img, *.dd, *.raw, *.bin)
Raw Split (For example: *.001, *.002, *.aa, *.ab, etc)
EnCase (For example: *.e01, *.e02, etc)
Virtual Machines (For example: *.vmdk, *.vhd)

4.1.3 Keyword Search Module:

The Keyword Search module facilitates both the ingest portion of searching and also supports manual text searching after ingest has completed. It extracts text from files being ingested, selected reports generated by other modules, and results generated by other modules. This extracted text is then added to a Solr index that can then be searched.

Autopsy tries its best to extract the maximum amount of text from the files being indexed. First, the indexing will try to extract text from supported file formats, such as pure text file format, MS Office Documents, PDF files, Email, and many others. If the file is not supported by the standard text extractor, Autopsy will fall back to a string extraction algorithm. String extraction on unknown file formats or arbitrary binary files can often extract a sizeable amount of text from a file, often enough to provide additional clues to reviewers. String extraction will not extract text strings from encrypted files.

Autopsy ships with some built-in lists that define regular expressions and enable the user to search for Phone Numbers, IP addresses, URLs and E-mail addresses. However, enabling some of these very general lists can produce a very large number of hits, and many of them can be false-positives. Regular expressions can potentially take a long time to complete.

Once files are placed in the Solr index, they can be searched quickly for specific keywords, regular expressions, or keyword search lists that can contain a mixture of keywords and regular expressions. Search queries can be executed automatically during the ingest run or at the end of the ingest, depending on the current settings and the time it takes to ingest the image.

3.1.4 File Search:

The File Search tool can be accessed either from the Tools menu or by right-clicking on a data source node in the Data Explorer / Directory Tree. By using File Search, you can specify, filter, and show the directories and files that you want to see from the images in the currently opened case. The File Search results will be populated in a brand new Table Result viewer on the right-hand side.

There are several categories that you can use to filter and show the directories and files within the images in the current opened case. The categories are:

- ✓ **Name**: Search for all files and directory whose name contains the pattern given.
- ✓ **Size**: Search for all files and directory whose size matches the pattern given. The pattern can be "equal to", "greater than", and "less than". The unit for the size can be "Byte(s)", "KB", "MB", "GB", and "TB".
- ✓ **MIME (Multipurpose Internet Mail Extension) Type**: Search for all files with the selected MIME type.
- ✓ **MD5**: Search for all files with the given MD5 hash.

- ✓ **Date**: Search for all files and directory whose "date property" is within the date range given. The "date properties" are "Modified Date", "Accessed Date", "Changed Date", and "Created Date". You must also specify the time zone for the date given.
- ✓ **Known Status**: Search for all files and directory whose known status is recognized as either Unknown, Known, or Notable.
- ✓ **Data Source**: Search only within the specified data source instead of the entire case.

4.1.5 PhotoRec Carver Module:

The PhotoRec Carver module carves files from unallocated space in the data source and sends the files found through the ingest processing chain. This can help a reviewer discover more information about files that used to be on the device and were subsequently deleted. These are simply extra files that were found in "empty" portions of the device storage

4.1.6 Hash Lookup Module:

The **Hash Lookup Module** calculates MD5 hash values for files and looks up hash values in a database to determine if the file is notable, known (in general), or unknown. **Hash sets** are used to identify files that are 'known' or 'notable'. **Known good files** are those that can be safely ignored. This set of files frequently includes standard OS and application files. Ignoring such uninteresting-to-the-investigator files, can greatly reduce image analysis time. **Notable (or known bad) files** are those that should raise awareness. This set will vary depending on the type of investigation, but common examples include contraband images and malware.

4.1.7 Tagging:

Tagging (or Bookmarking) allows you to create a reference to a file or object and easily find it later or include it in a report. Tagging is also used by the central repository to mark items as notable.

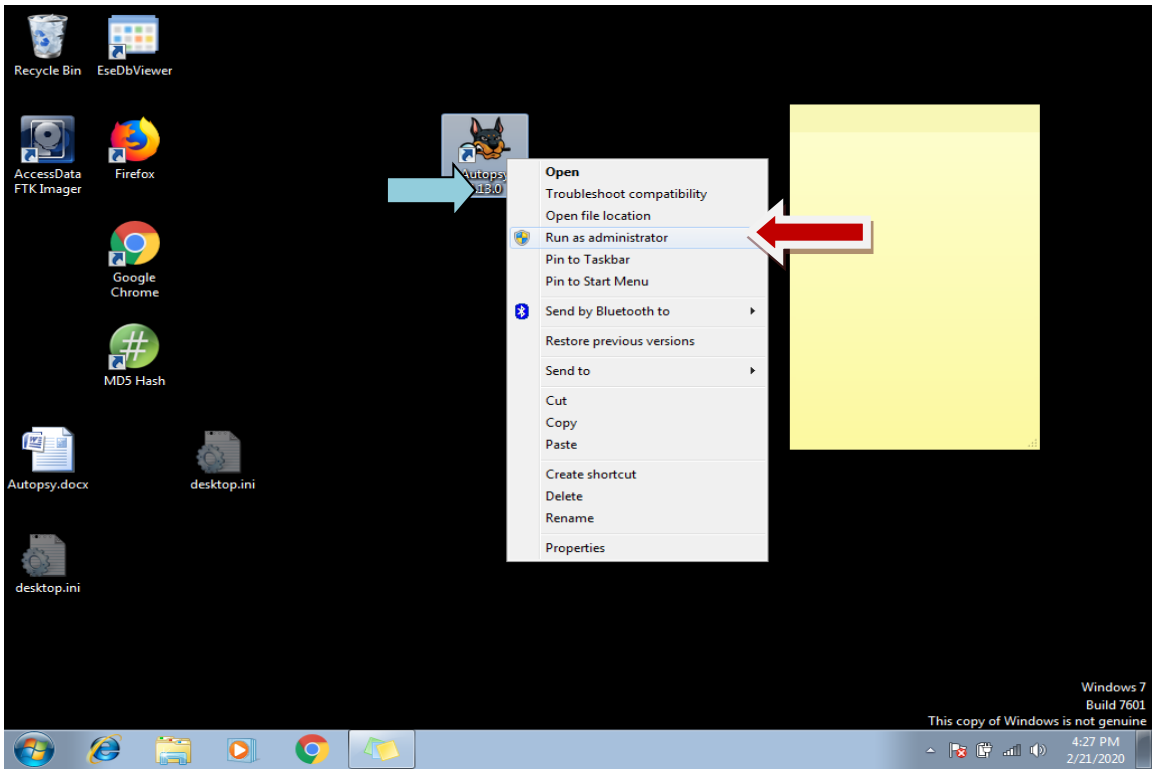
4.1.8 Report module:

The report modules allow the user to extract key information from a case in a variety of formats. This includes making an HTML or Excel report containing all the extracted content, keyword hits, etc. from a case, or creating a KML file out of any coordinates found to load into software like Google Earth.

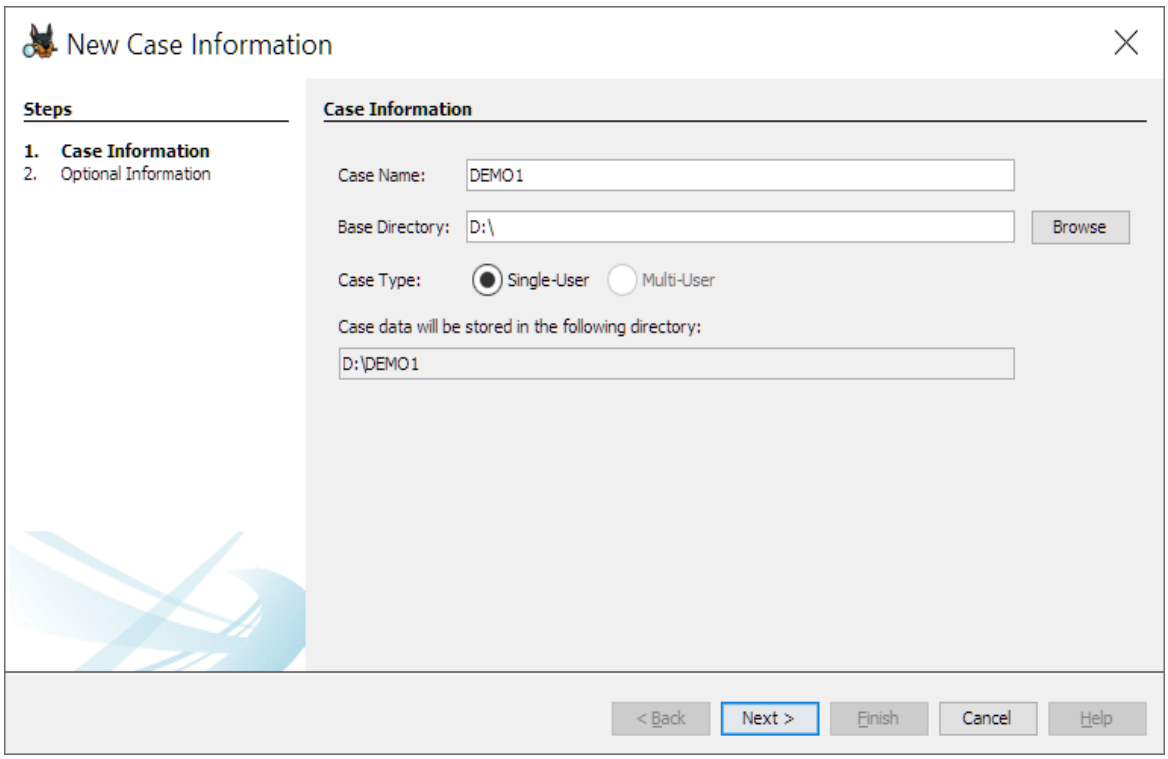
4.2 Instructions

How to use Autopsy to analyze data?

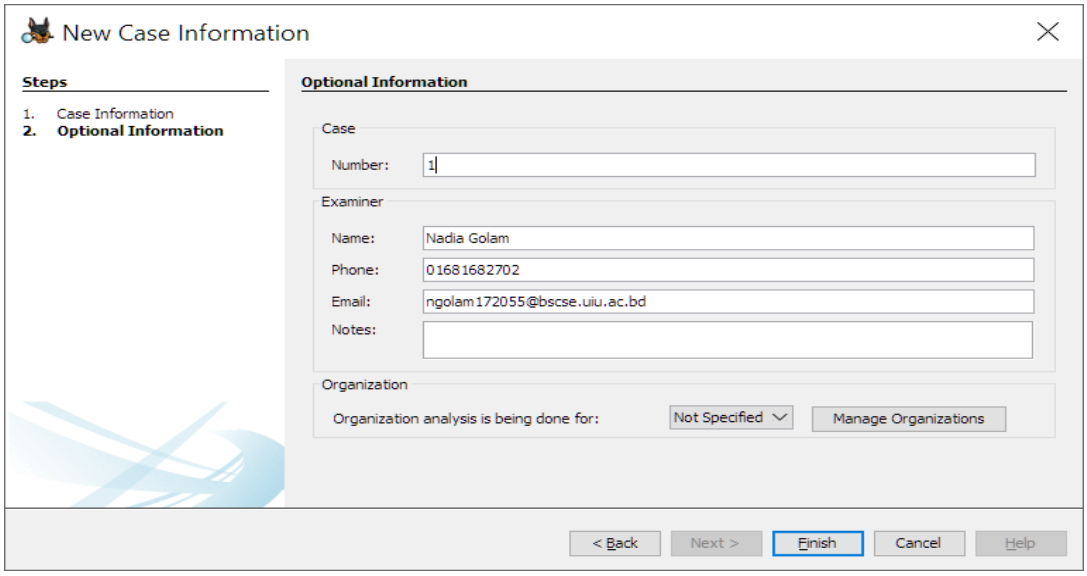
1. Run "Autopsy" as an administrator from your PC.



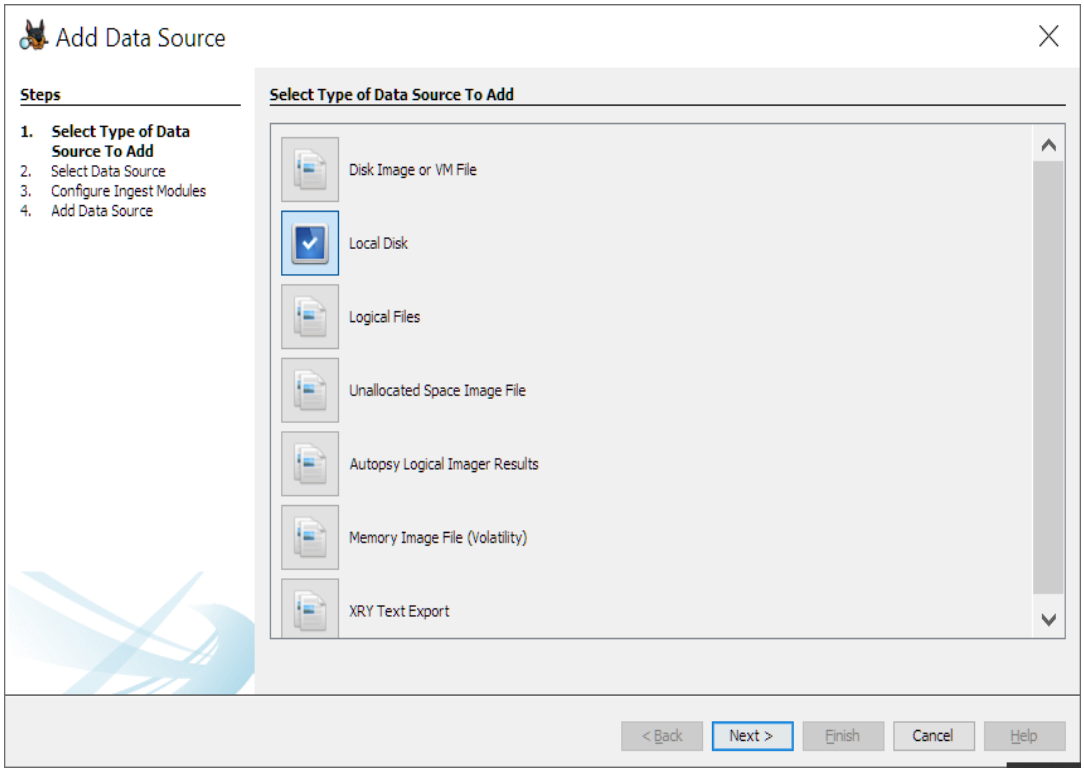
2. Open a new case. Fill up the case information giving a name to the case, select a directory to keep that and select the type of case to single-user as only one person is investigating. We can also use Multi-User, then multiple investigators can access in this particular case. Click “Next”.

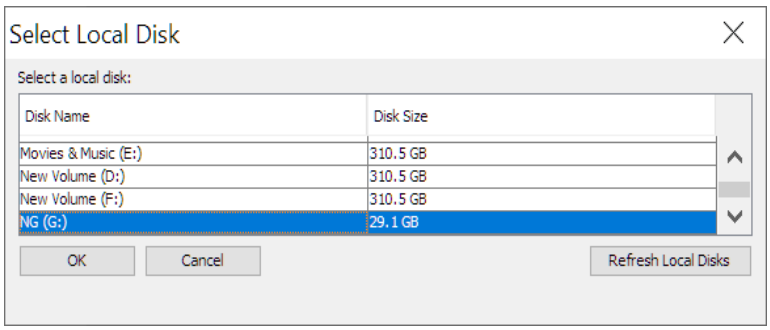


3. Give case number and examiner information then click “Finish”.

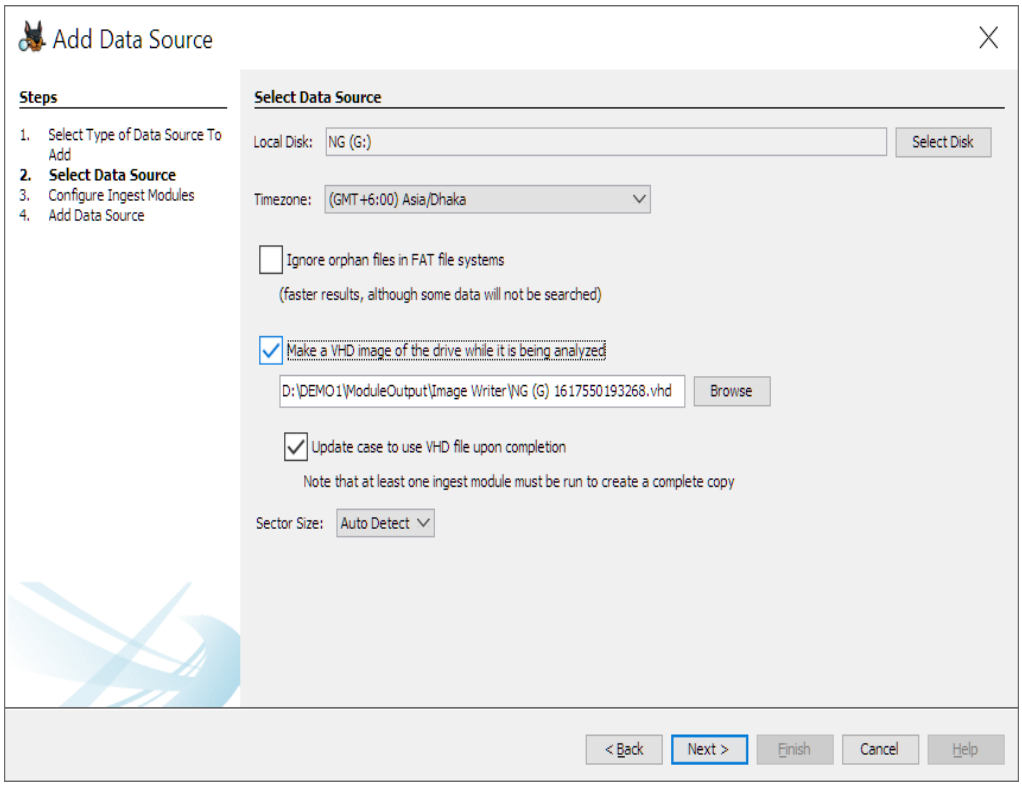


4. Add data source, you can use a local disk i.e. a pen drive to examine.

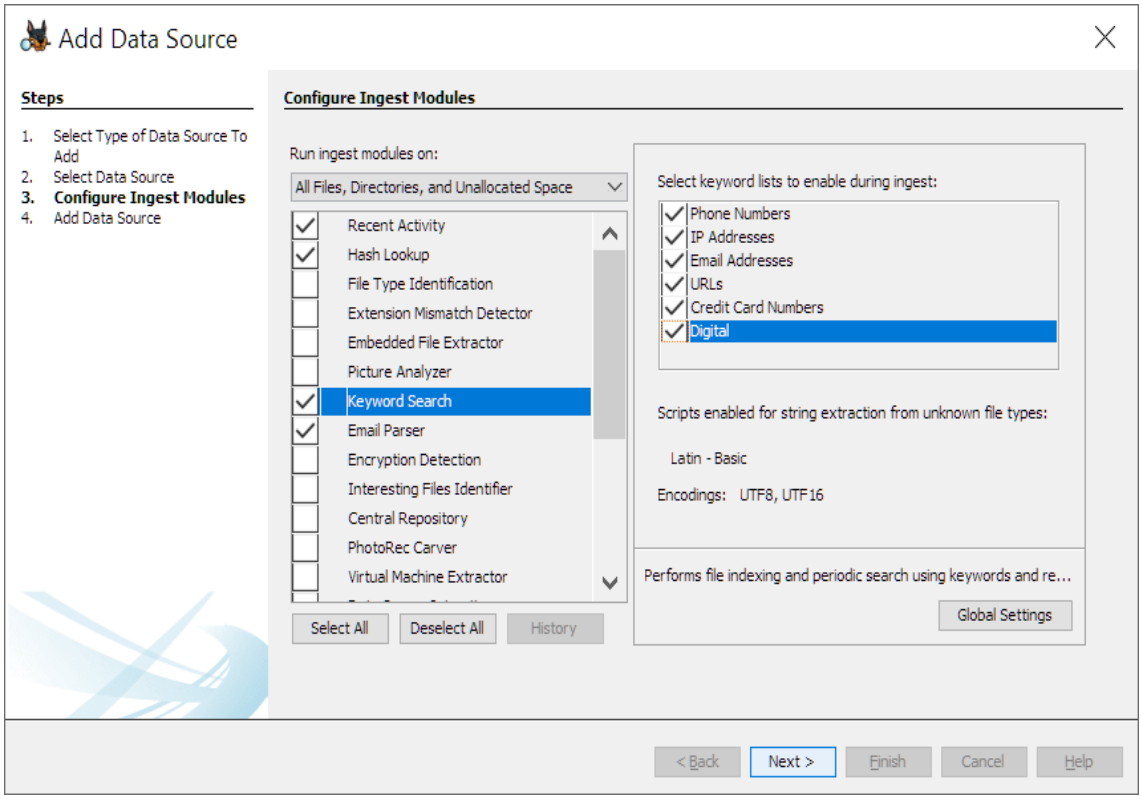




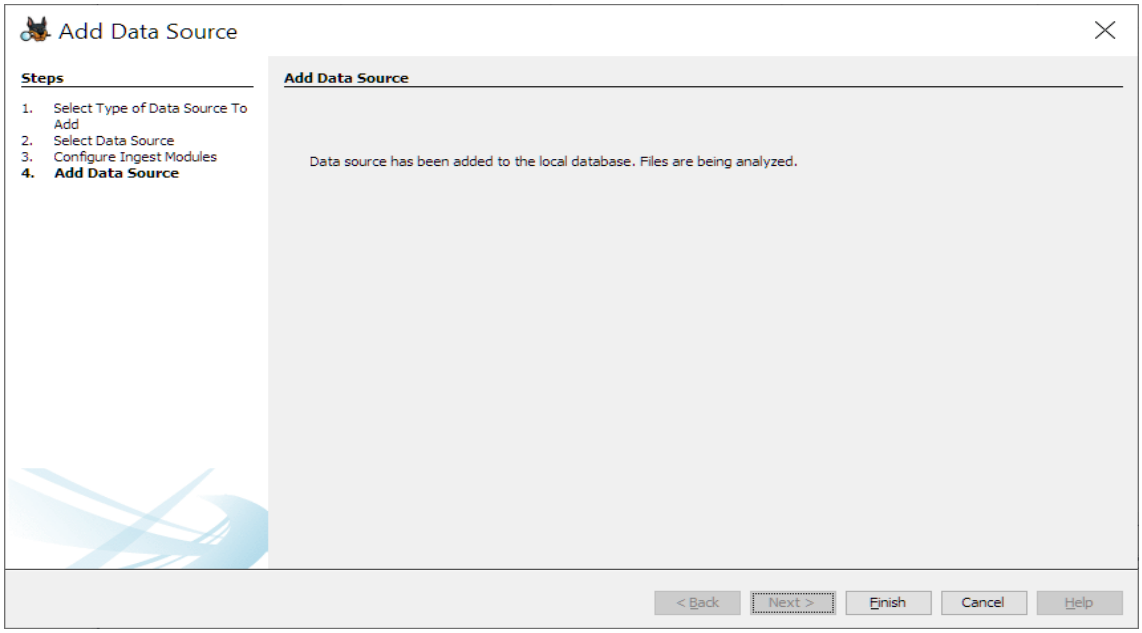
5. After that data source is selected, check the VHD image of the drive, so that a virtual hard disk of the pen drive is created while it is being analyzed.



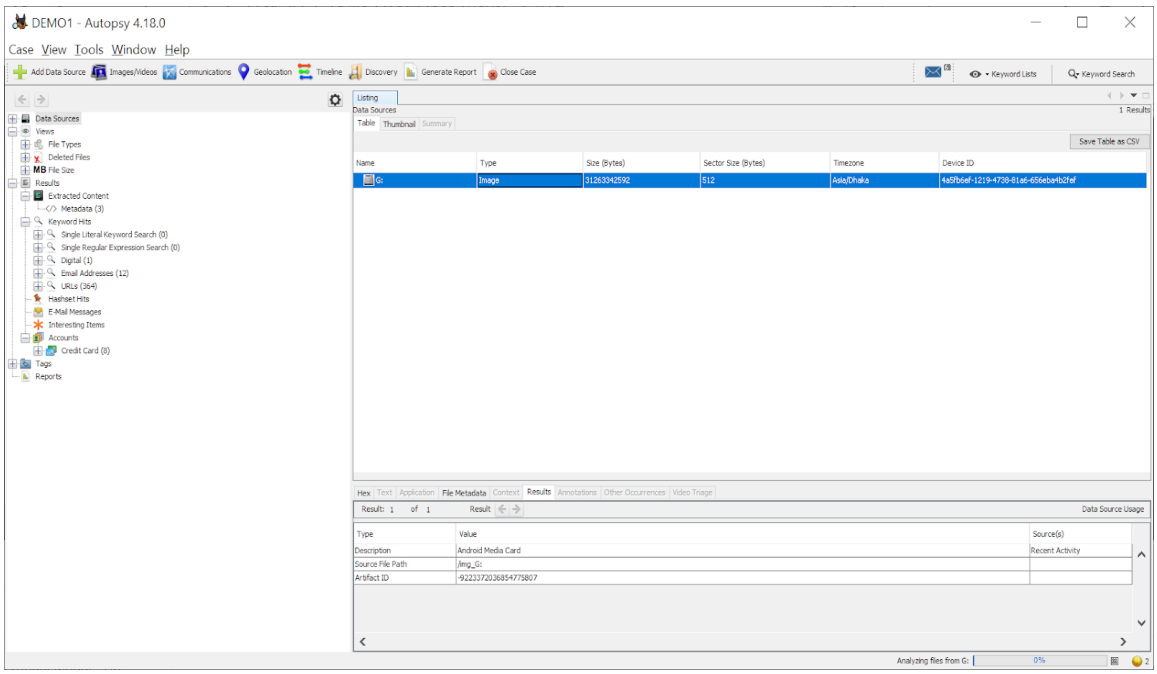
6. Ingest modules are selected accordingly, here in this case we have used the modules recent activity module which will extract any information related to the web and also operating system, hash lookup, keyword search and email parser.



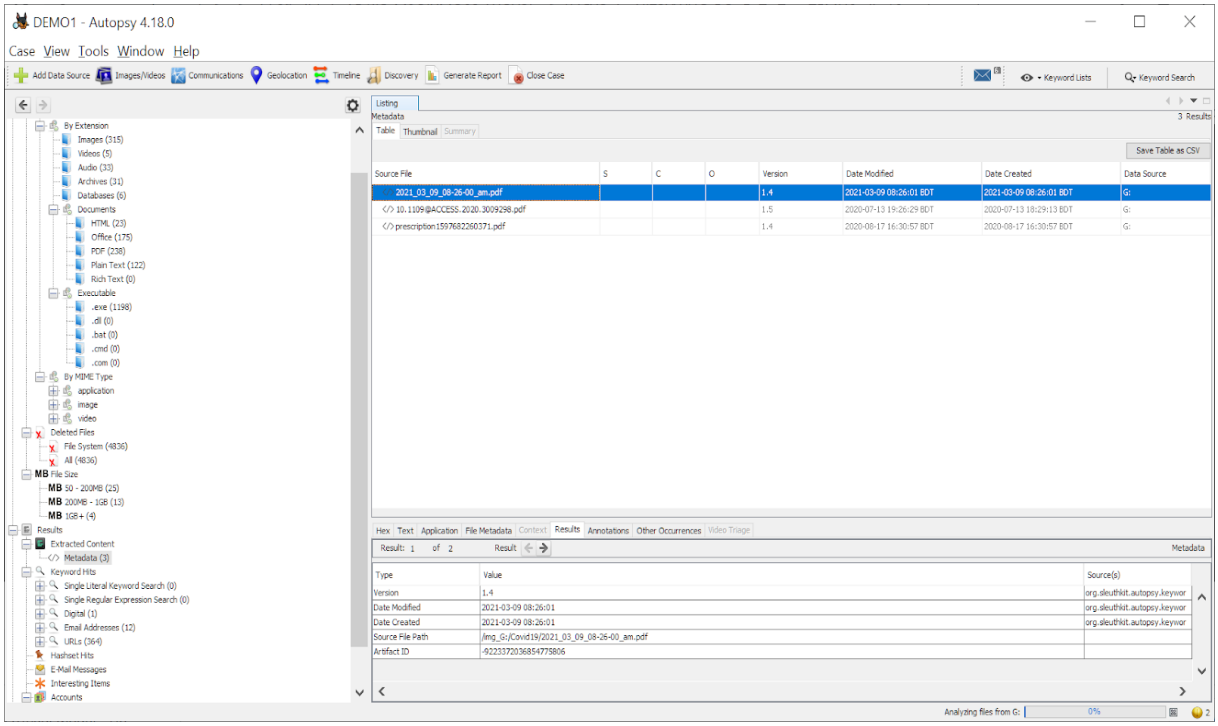
7. The next window appears, click the “finish” button.



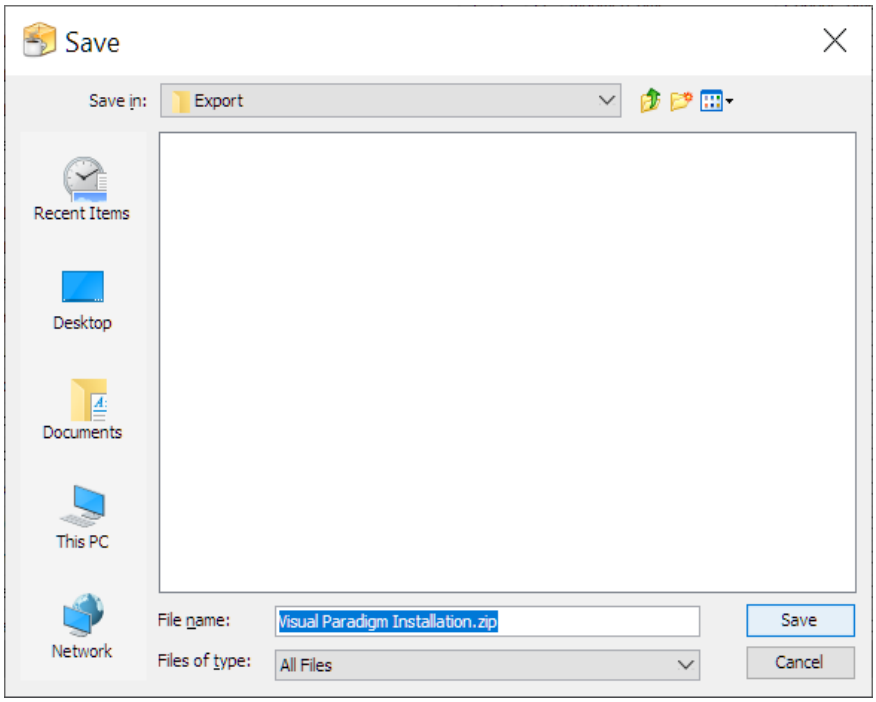
A window appears as the following figure.

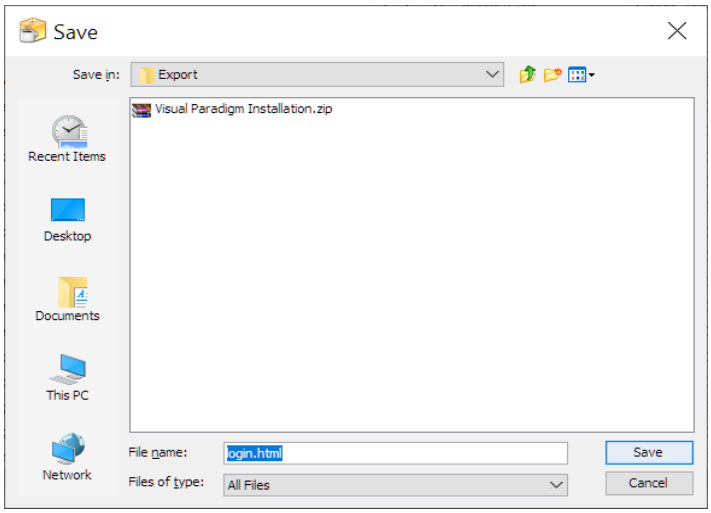


8. In the metadata, three files with corresponding information are shown. Expand the file types section and you will get images, videos, audio, archive and database extension. Documents and executable section also lies under file types section.

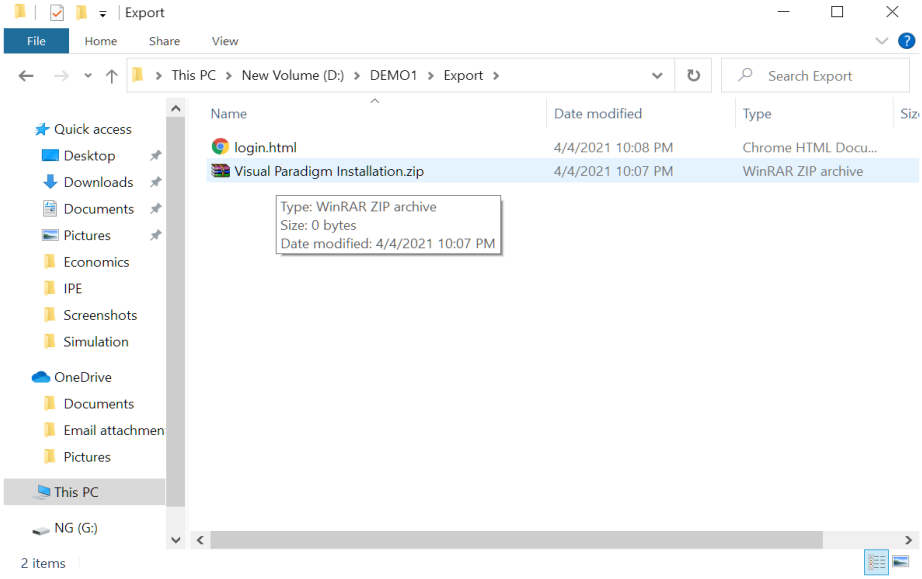


9. Let us extract a zip file and an html file.

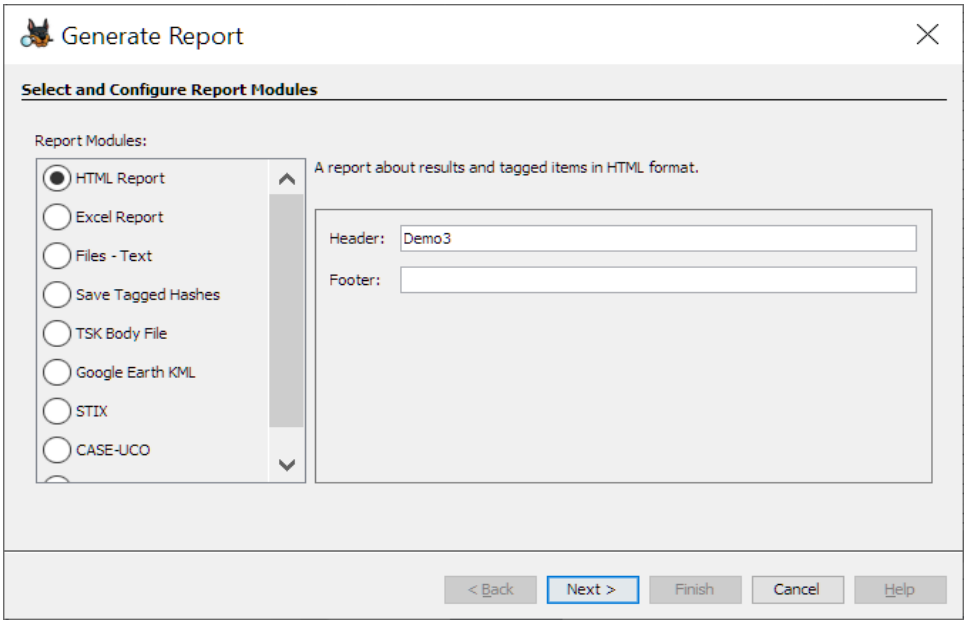




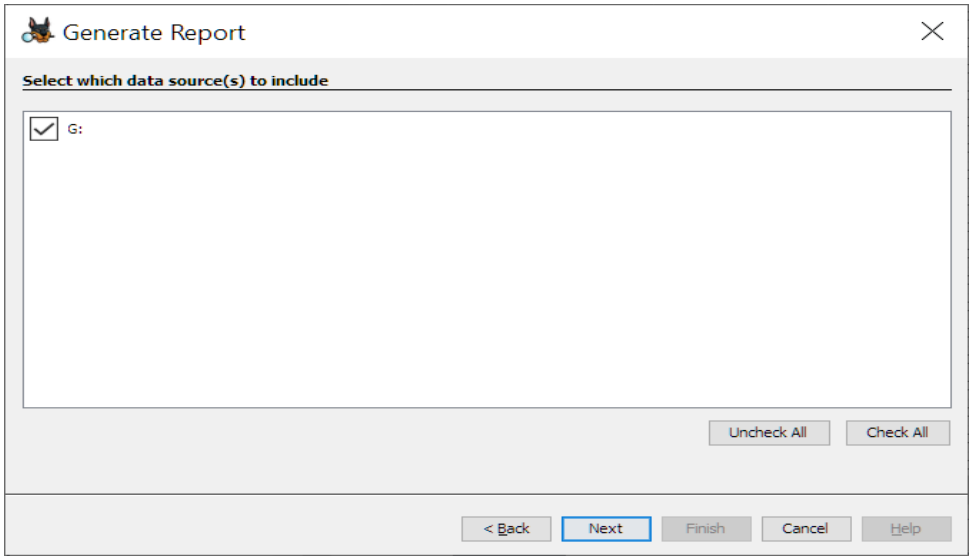
The extracted files are then found in the export directory.



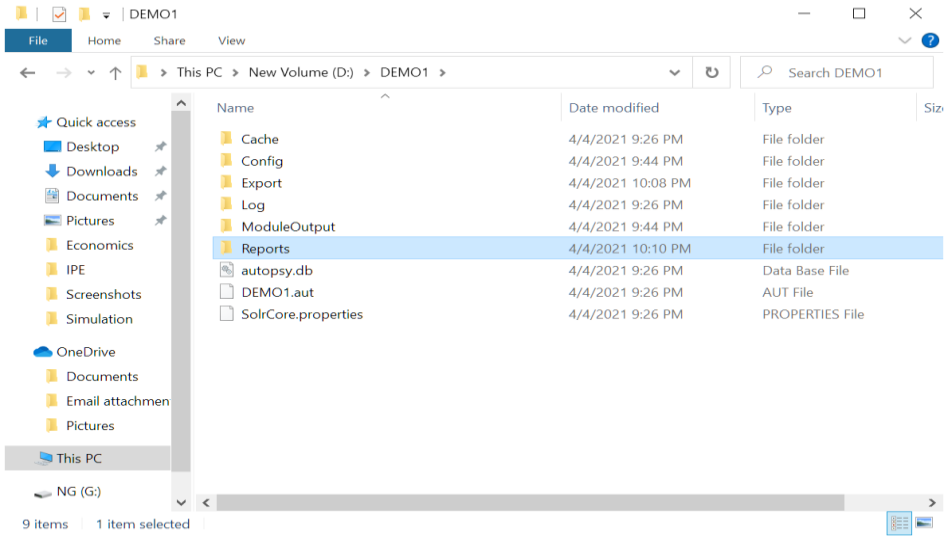
10. Generate a report of the case.

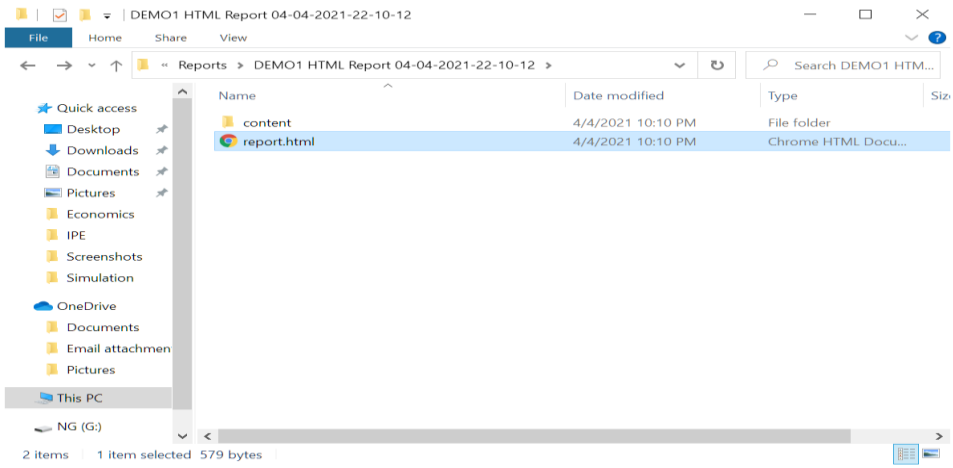


G drive is selected as a data source to generate a report.

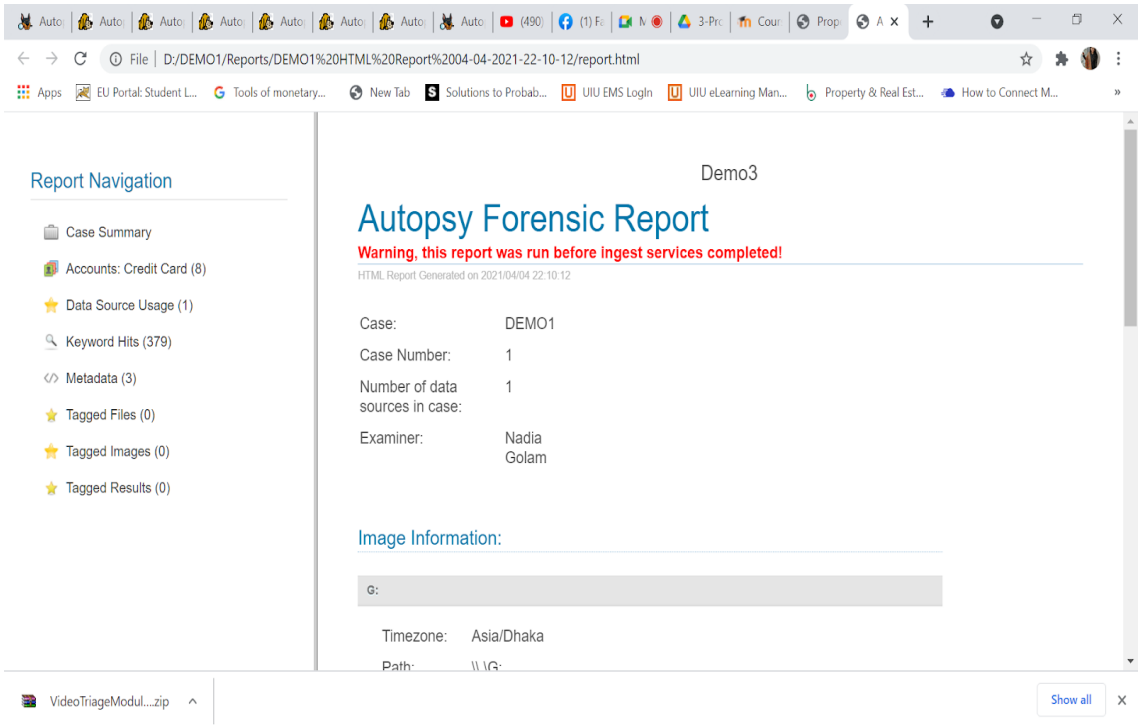


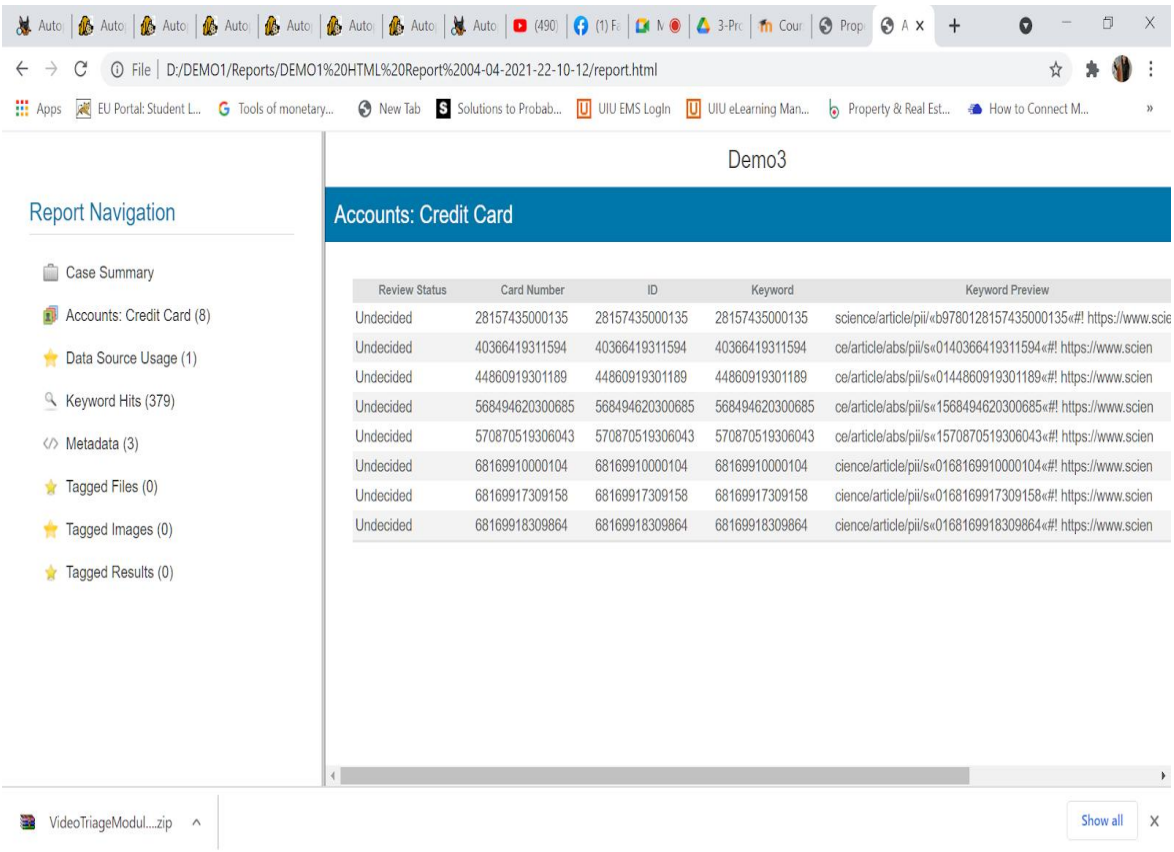
The report is found in the Reports directory and the file is kept as an html file.

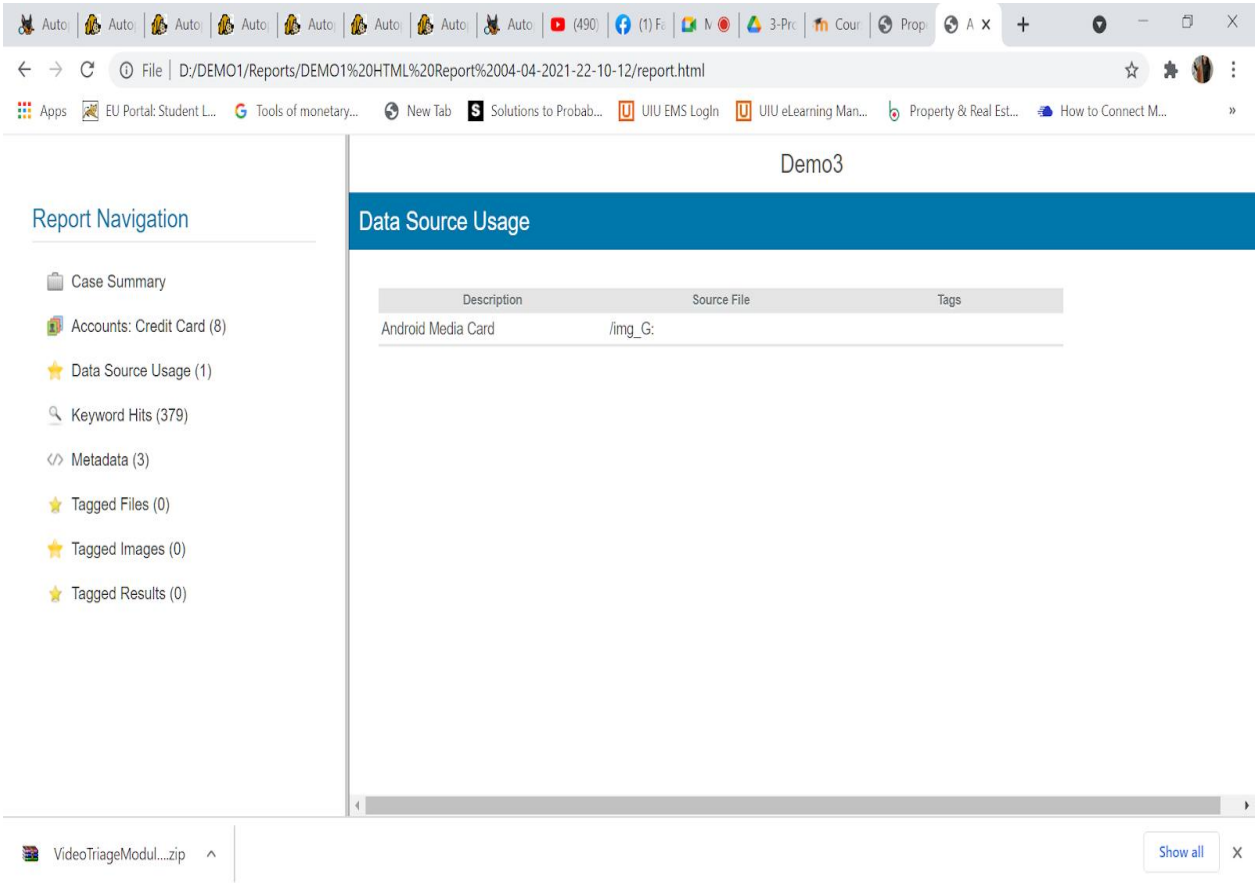


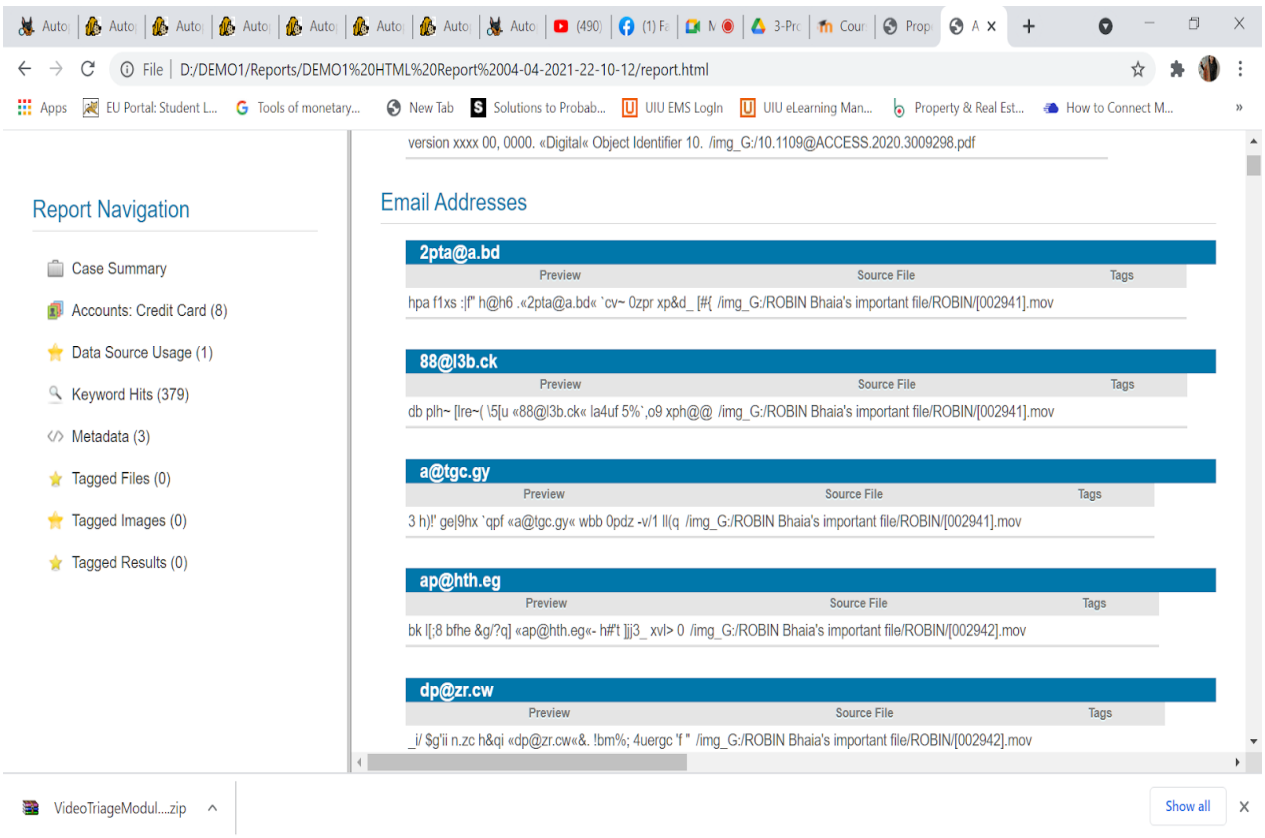


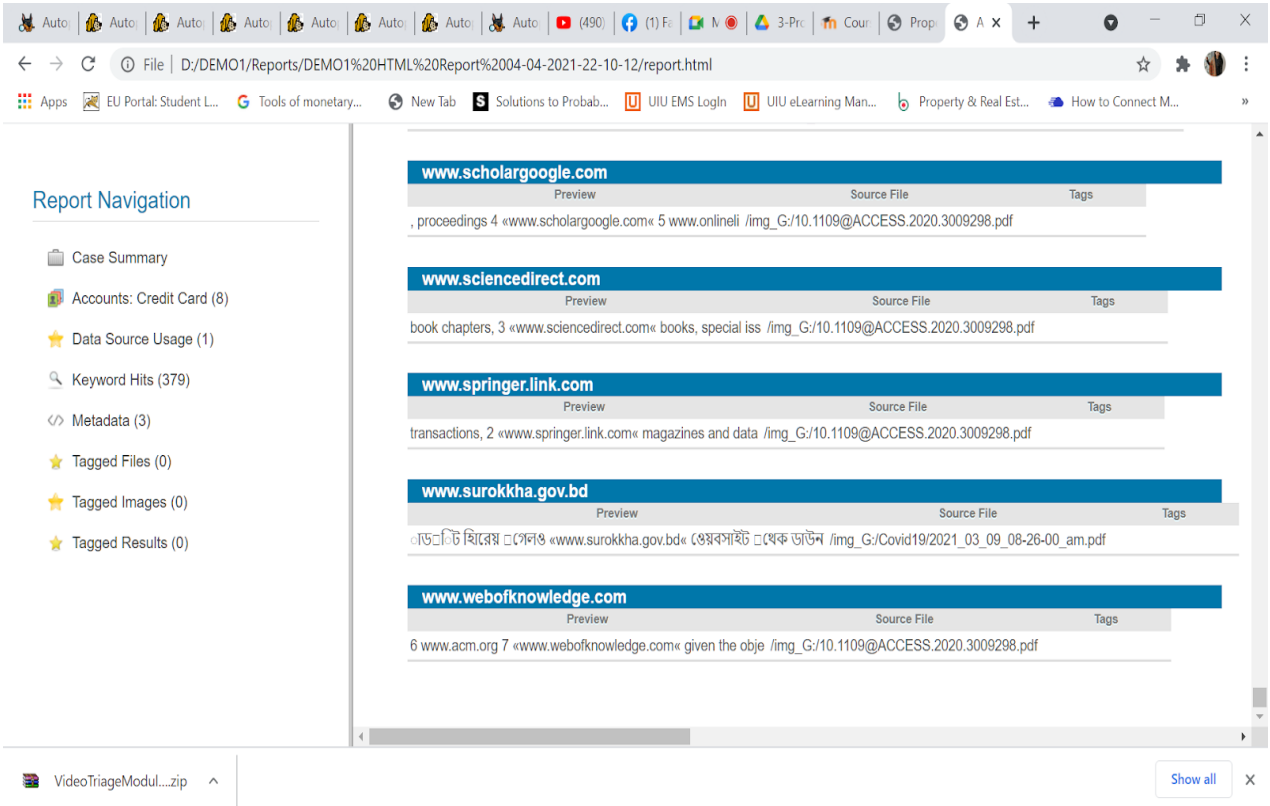
After clicking on the html file, the report appears as shown in the following figure.





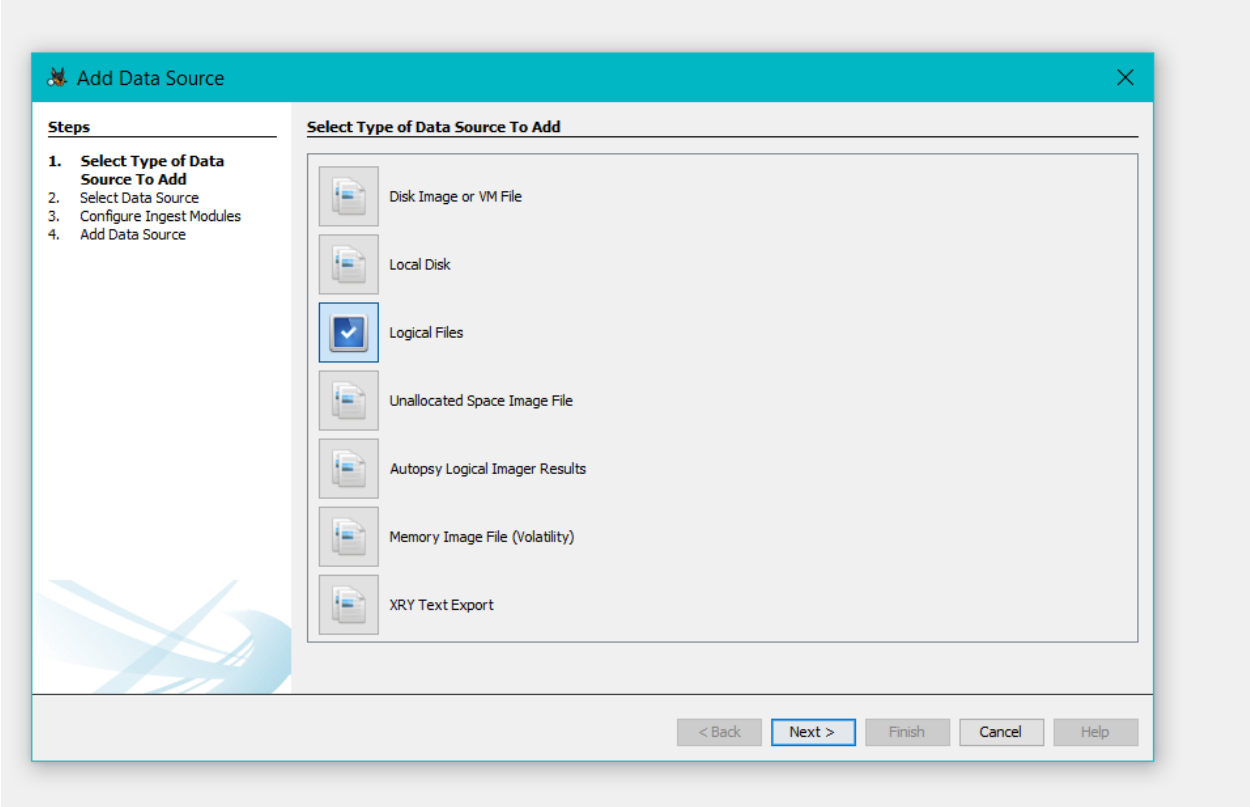

File | D:/DEMO1/Reports/DEMO1%20HTML%20Report%2004-04-2021-22-10-12/report.html
Demo3
Report Navigation
Case Summary
Accounts: Credit Card (8)
Data Source Usage (1)
Keyword Hits (379)
Metadata (3)
Tagged Files (0)
Tagged Images (0)
Tagged Results (0)
Data Source Usage
Description
Source File
Tags
Android Media Card
/img_G:
VideoTriageModul....zip
Show all


File | D:/DEMO1/Reports/DEMO1%20HTML%20Report%2004-04-2021-22-10-12/report.html
version xxxx 00, 0000. «Digital« Object Identifier 10. /img_G:/10.1109@ACCESS.2020.3009298.pdf
Report Navigation
Case Summary
Accounts: Credit Card (8)
Data Source Usage (1)
Keyword Hits (379)
Metadata (3)
Tagged Files (0)
Tagged Images (0)
Tagged Results (0)
Email Addresses
2pta@a.bd
Preview
Source File
Tags
hpa f1xs :|f" h@h6 .«2pta@a.bd« `cv~ 0zpr xp&d_ [#{ /img_G:/ROBIN Bhaia's important file/ROBIN/[002941].mov
88@l3b.ck
db plh~ [lre~(\5[u «88@l3b.ck« la4uf 5%`,o9 xph@@ /img_G:/ROBIN Bhaia's important file/ROBIN/[002941].mov
a@tgc.gy
3 h)!' ge|9hx `qpf «a@tgc.gy« wbb 0pdz -v/1 ll(q /img_G:/ROBIN Bhaia's important file/ROBIN/[002941].mov
ap@hth.eg
bk l[;8 bfhe &g/?q] «ap@hth.eg«- h#'t]jj3_ xvl> 0 /img_G:/ROBIN Bhaia's important file/ROBIN/[002942].mov
dp@zr.cw
_i/ $g'ii n.zc h&qi «dp@zr.cw«&. !bm%; 4uergc 'f " /img_G:/ROBIN Bhaia's important file/ROBIN/[002942].mov
VideoTriageModul....zip
Show all

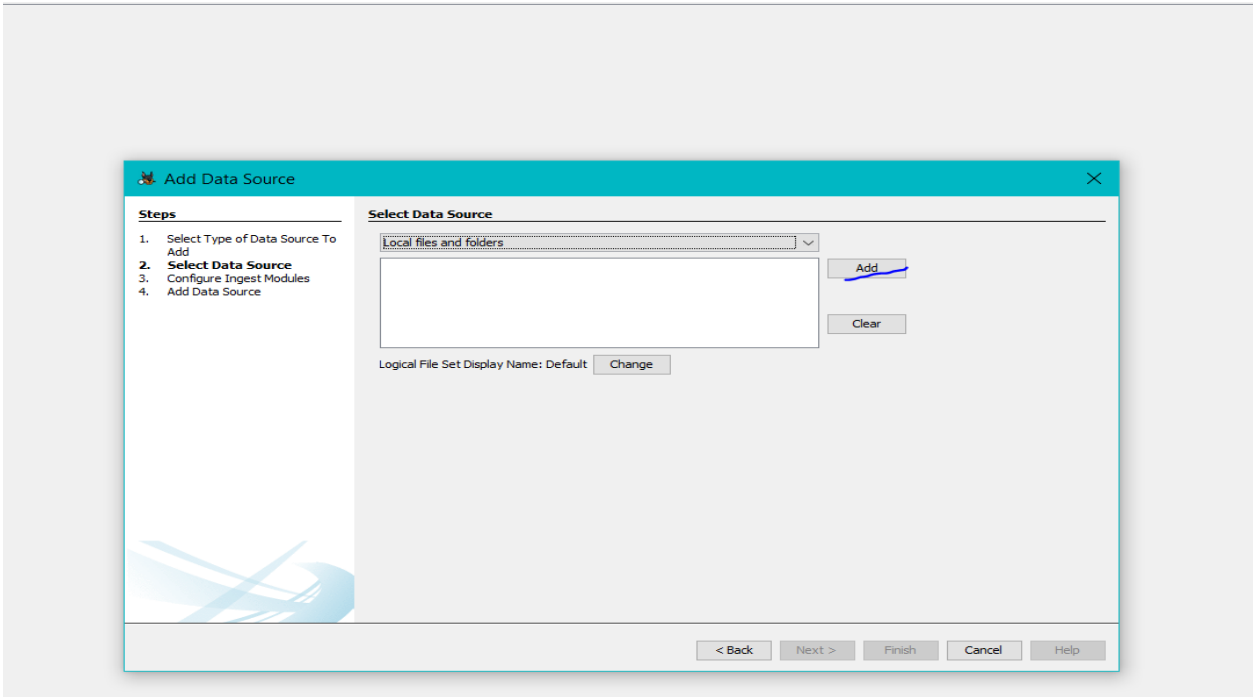


In the report, all the information about credit card, data source usage, keywords i.e. email and urls all are extracted.

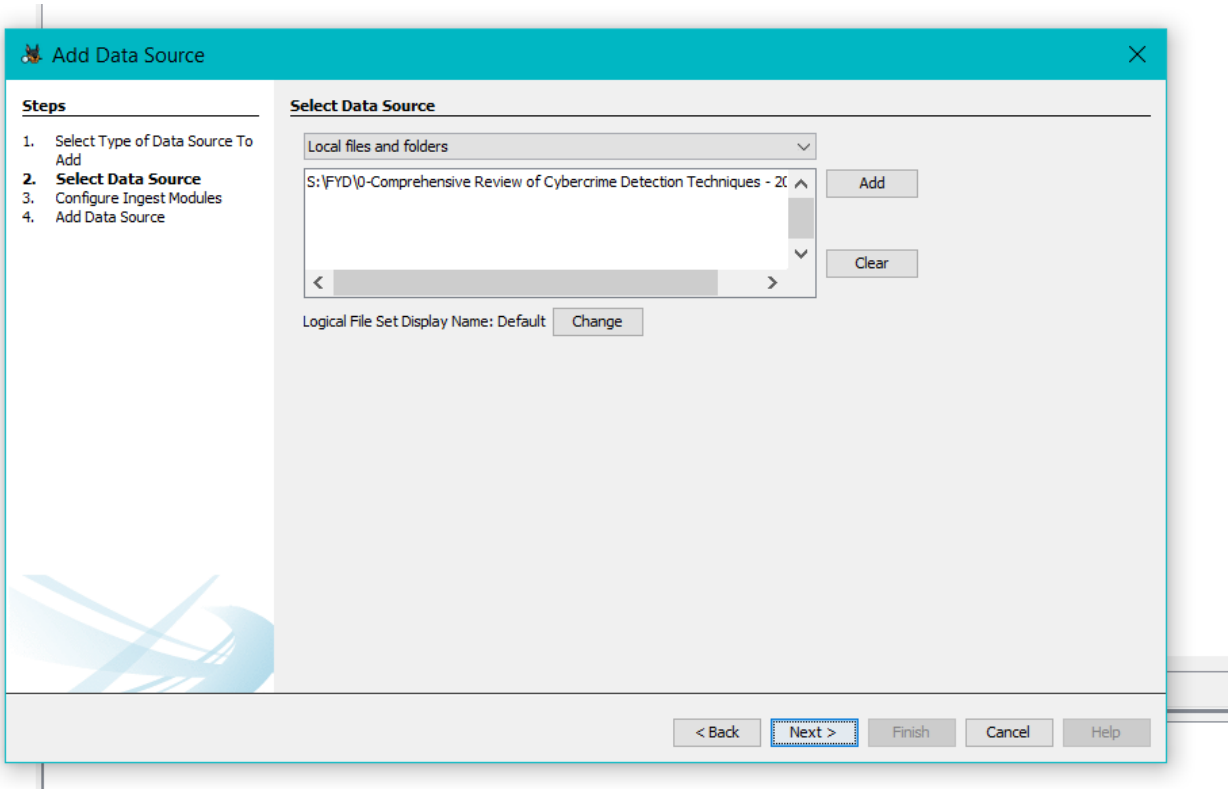
11. Some experiments are done using logical files in Autopsy. First, we have to select the type of data source to add. Here our data source is “Logical Files”.



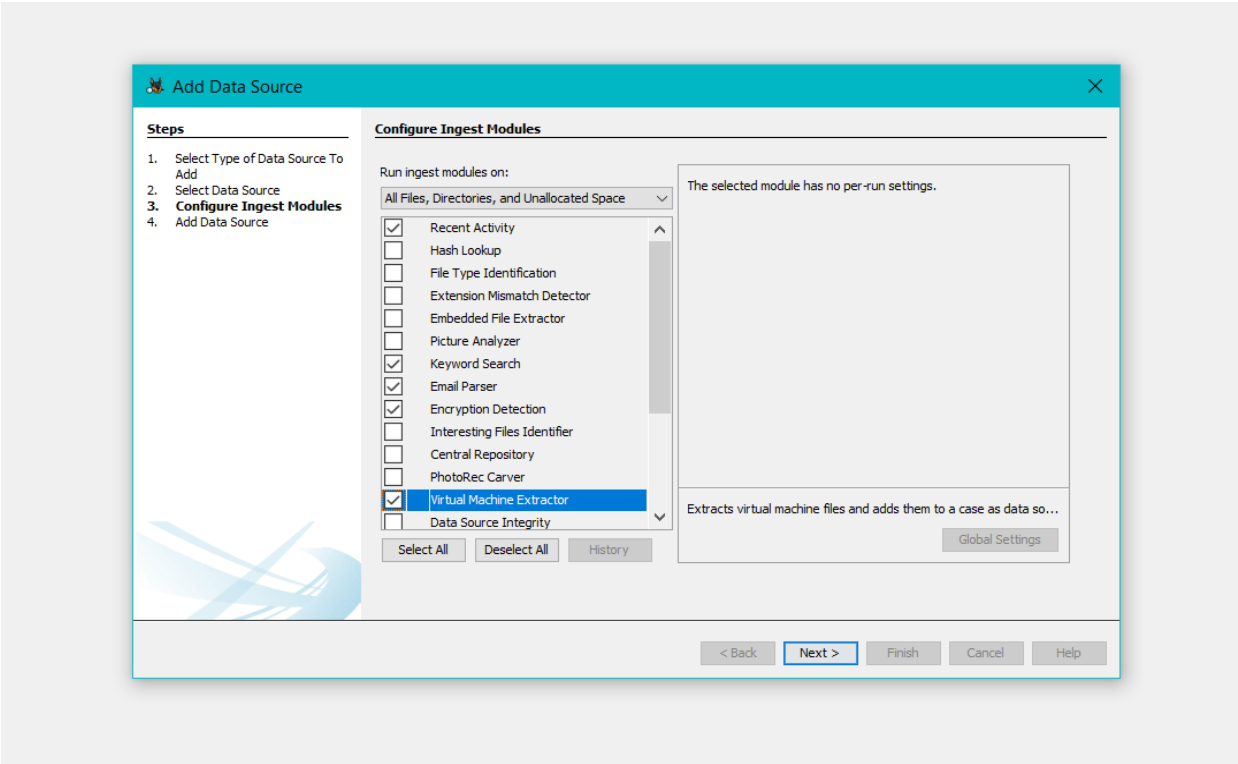
Add logical files which we need to analyze for our case.

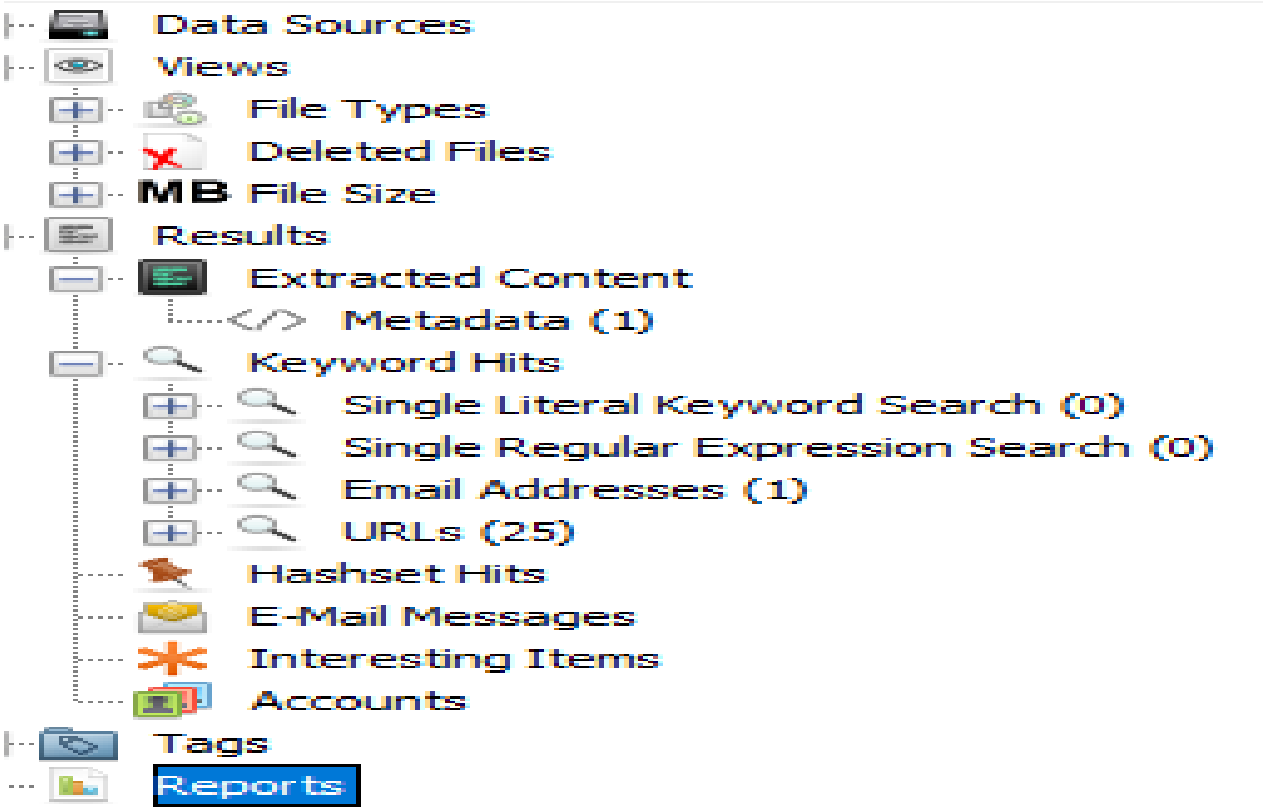


Add files which we need to investigate.

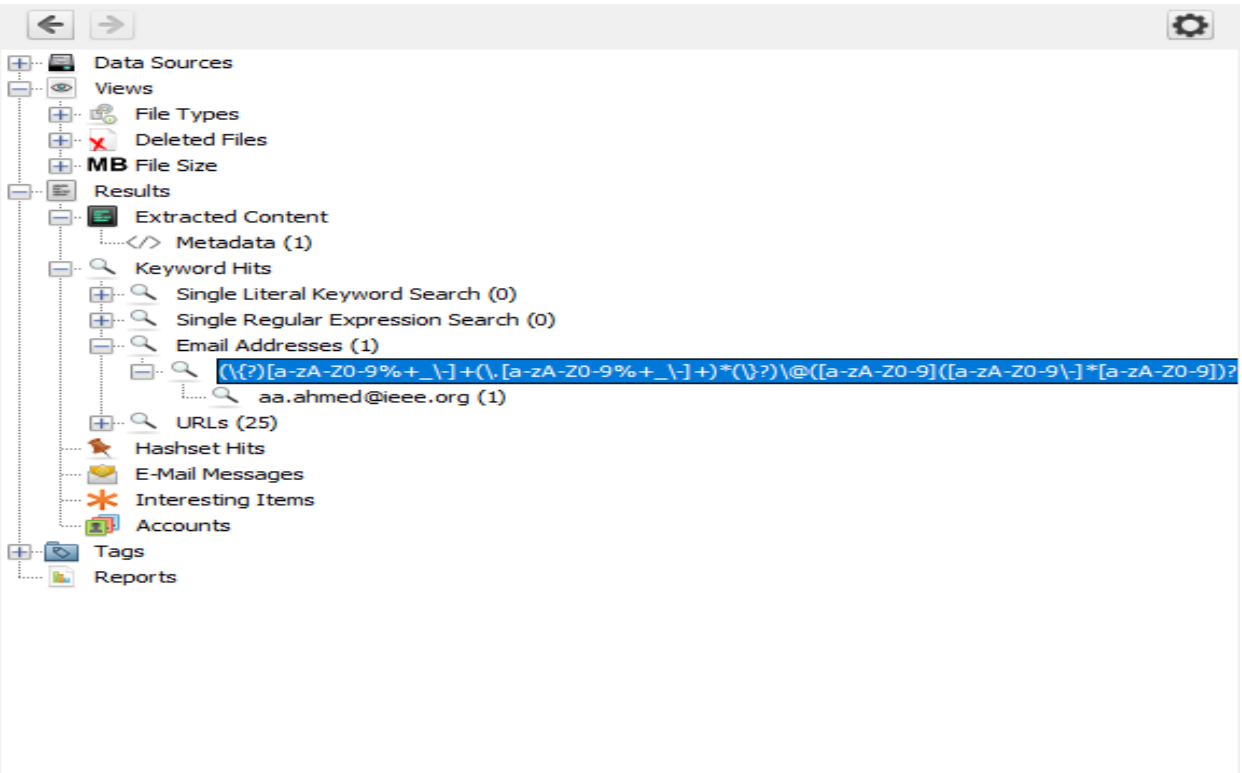


12. In this section named "Configure Ingest Modules" we have to select "Recent Activity", "Keyword Search", "Email Parser", "Encryption Detection", "Virtual Machine Extractor". Click on "Results".

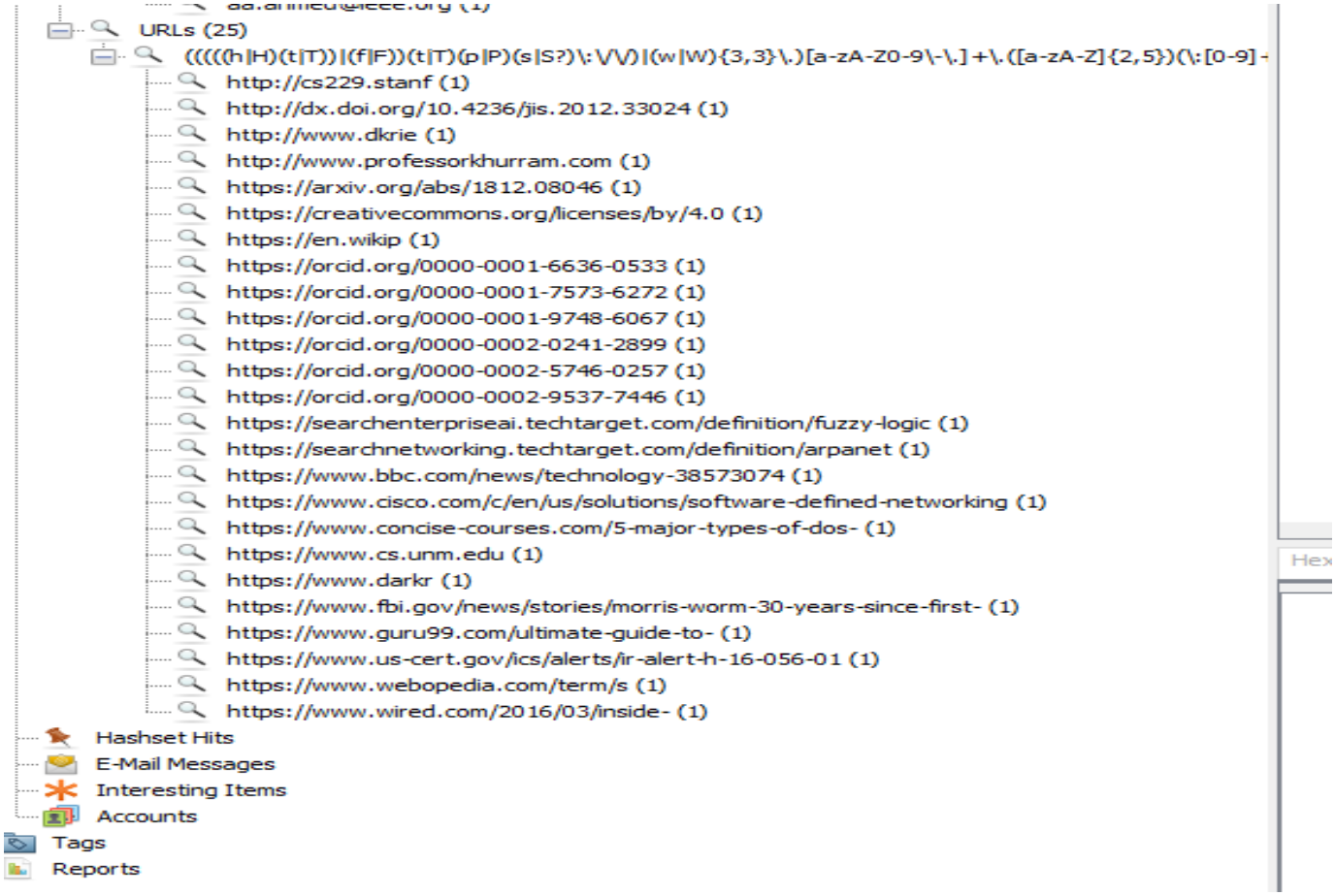




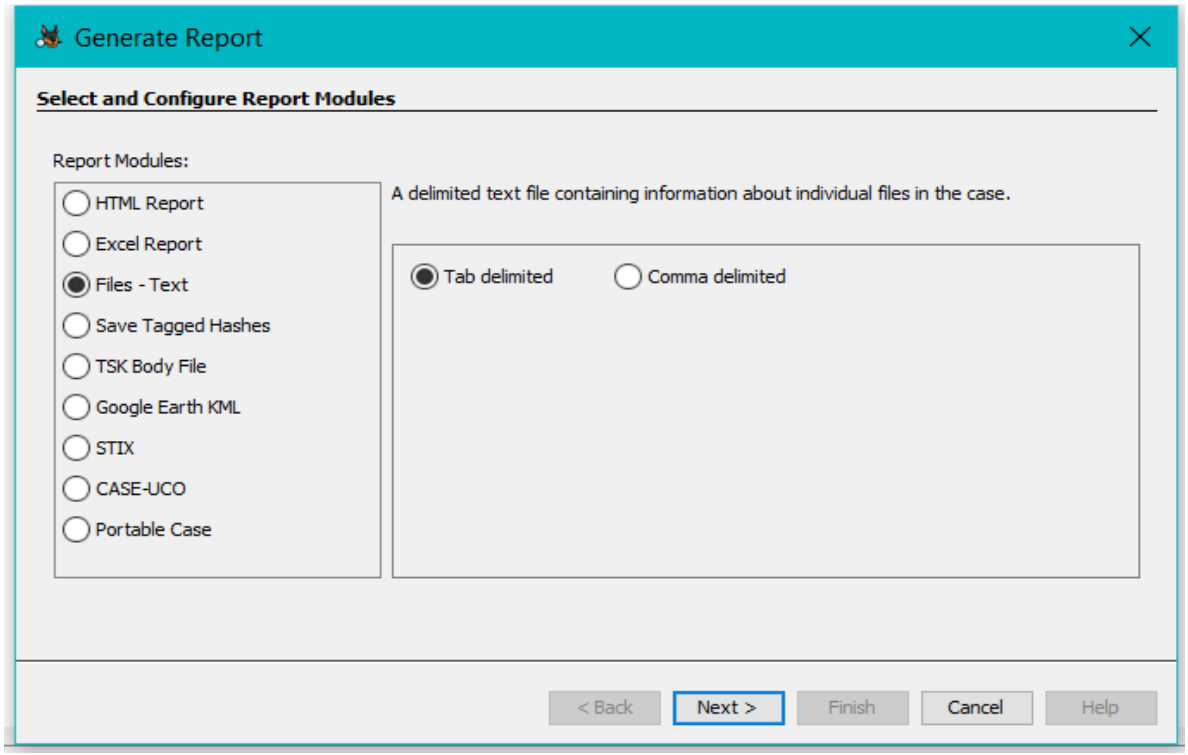
Emails are shown here.



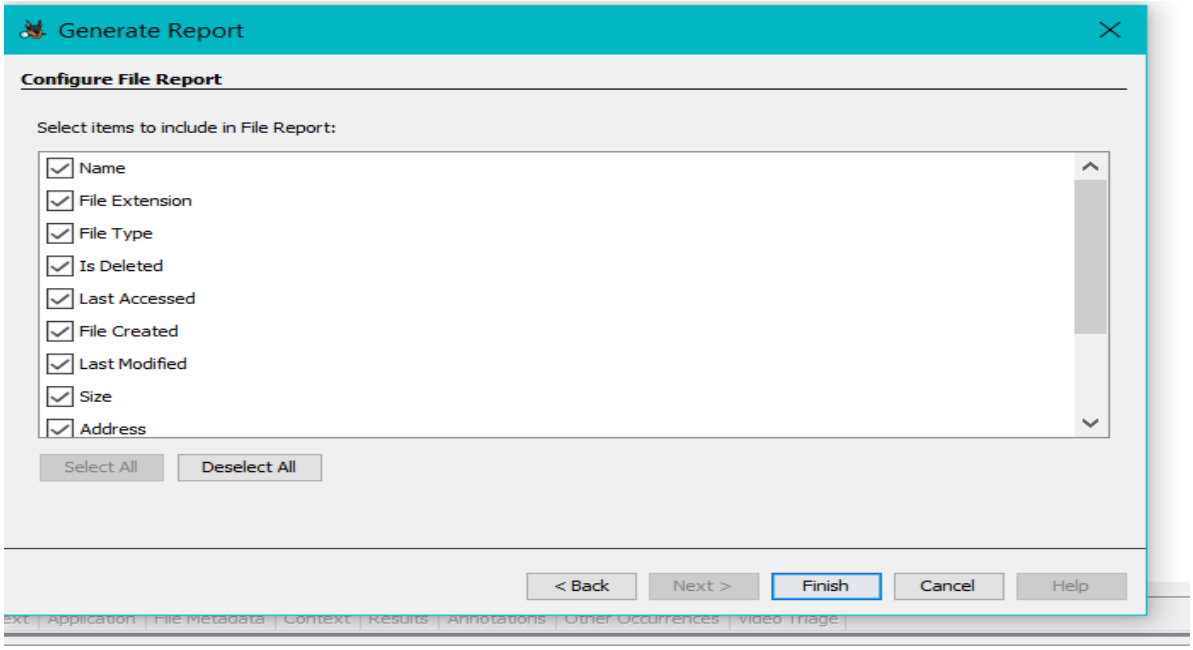
URLs are shown here.



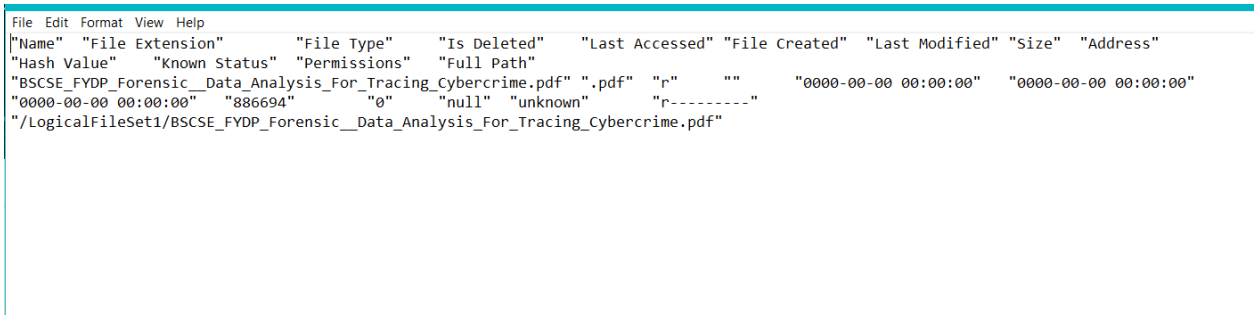
13. Select “Files - Text” from Report Modules and “Tab delimited”.



14. Configure file report.

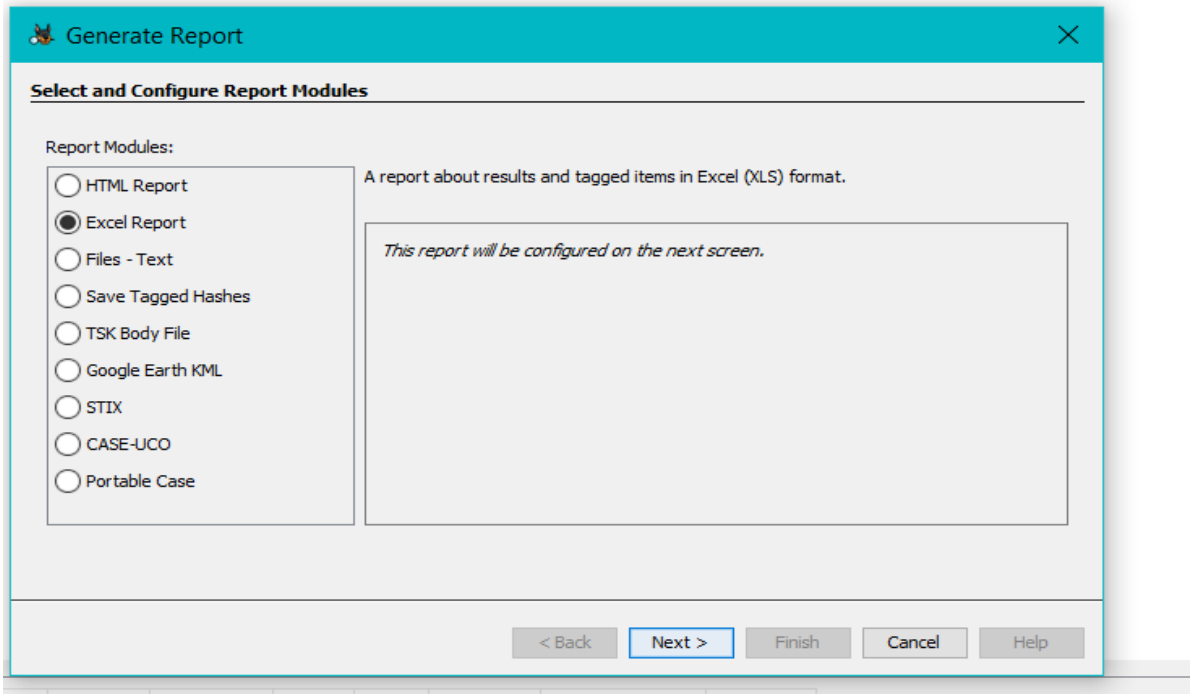


Report of the text file.

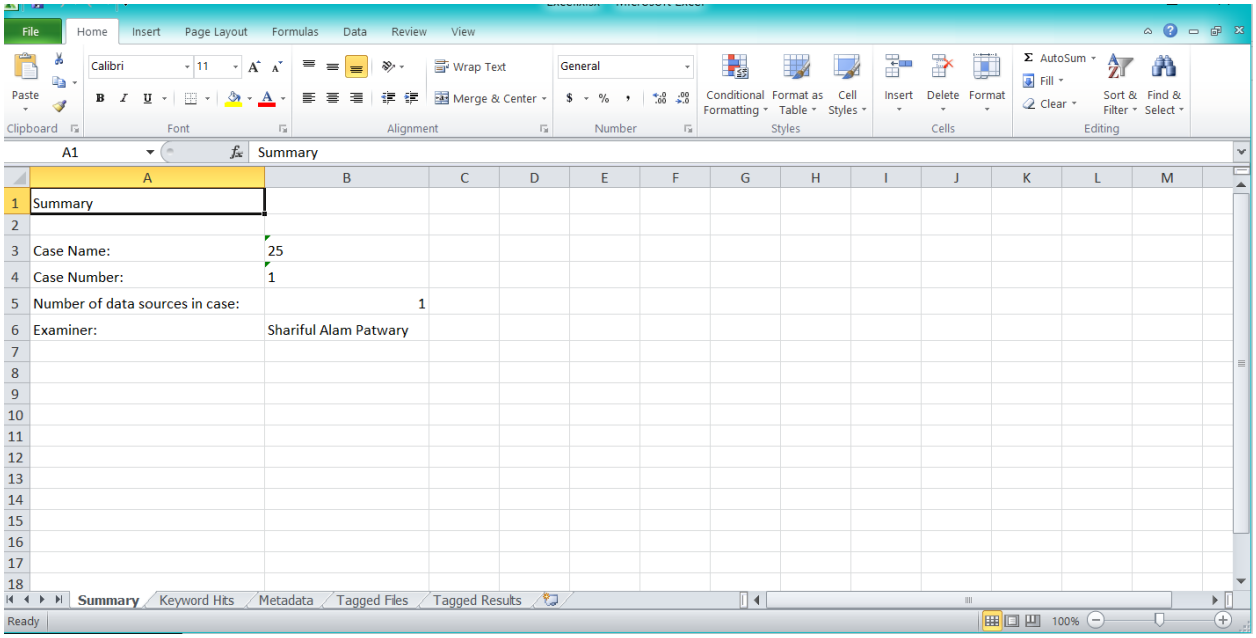


```
File  Edit  Format  View  Help
"Name"  "File Extension"        "File Type"     "Is Deleted"    "Last Accessed" "File Created"  "Last Modified" "Size"  "Address"
"Hash Value"    "Known Status"  "Permissions"   "Full Path"
"BSCSE_FYDP_Forensic__Data_Analysis_For_Tracing_Cybercrime.pdf" ".pdf"  "r"     ""      "0000-00-00 00:00:00"   "0000-00-00 00:00:00"
"0000-00-00 00:00:00"   "886694"        "0"     "null"  "unknown"       "r---------"
"/LogicalFileSet1/BSCSE_FYDP_Forensic__Data_Analysis_For_Tracing_Cybercrime.pdf"
```

15. Select “Excel Report”.



Report of Excel File.



	A	B
1	Summary	
2		
3	Case Name:	25
4	Case Number:	1
5	Number of data sources in case:	1
6	Examiner:	Shariful Alam Patwary